

Article

The Use of PINN in Modeling of Thermoelectric Modules

Roman Kluger ¹, Ryszard Buchalik ^{2,*} and Iwona Nowak ¹

¹ Faculty of Applied Mathematics, Silesian University of Technology, 44-100 Gliwice, Poland; roman.kluger@polsl.pl (R.K.); iwona.nowak@polsl.pl (I.N.)

² Faculty of Energy and Environmental Engineering, Silesian University of Technology, 44-100 Gliwice, Poland

* Correspondence: ryszard.buchalik@polsl.pl

Abstract

This paper analyzes the potential application of Physics-Informed Neural Networks (PINNs) in solving equations that describe thermal–electrical processes in thermoelectric systems. Combining machine learning with the laws of physics, the PINN method can serve as an alternative to traditional numerical methods, particularly in the context of the miniaturization of cooling systems, heat pumps, and systems that convert thermal energy (heat flow) into electrical energy (e.g., heat recovery), as well as the implementation of models in embedded systems. The article presents a model of thermoelectric equations, explains how PINNs work, provides numerical results, and assesses the advantages and disadvantages of the proposed approach.

Keywords: thermoelectric module; physics-informed neural networks; thermal modeling; transient analysis; supercooling; heat transfer; heat pump; numerical simulation

1. Introduction

Thermoelectric modules, which utilize the Seebeck and Peltier effects, are widely used in refrigeration devices, temperature stabilization systems, and partial conversion of heat flow into electricity (e.g., waste heat recovery). Modeling their behavior requires solving strongly coupled equations that describe both heat conduction and thermoelectric phenomena whose parameters may depend on temperature.

Traditional numerical methods enable solving such equations, but they usually involve high computational costs, due to the need to discretize the domain. Analytical solutions, on the other hand, are only available for highly simplified models, e.g., assuming constant temperatures in areas where, in reality, thermal resistance occurs, leading to temperature gradients depending on the operating parameters of the system. This significantly limits their usefulness in the analysis of real systems.

Physics-Informed Neural Networks (PINN) are an alternative approach to solving differential equations that avoids explicit discretization of the domain. In this method, the neural network is trained to satisfy the equations that describe a given physical process (known as the governing equations). A key element of this approach is the use of automatic differentiation, a technique that enables efficient calculation of function derivatives already at the stage of determining the output values of the neural network. The mathematical foundations of automatic differentiation can be found, among others, in the works [1,2].

An important advantage of PINNs is their high flexibility, which allows for relatively easy inclusion of differential and algebraic equations, as well as initial and boundary conditions. This method was introduced in [3], while a Python library was proposed in [4] that allows the practical use of PINNs. That library called deepxde was used in this work.



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The article in [5] discusses selected theoretical issues related to the convergence of the algorithm and provides a broad overview of various PINN architectures.

The applications of PINNs have been studied in many areas of physics. An overview of their use in fluid mechanics is presented in [6], while applications in heat transfer problems are described in [7]. Examples of the use of PINNs in optimization problems can be found in [8], and an overview of applications in solid mechanics can be found in [9]. In the context of thermoelectric phenomena, however, the PINN method remains a relatively new and still poorly researched tool; previous attempts to apply it can be found, among others, in the works [10–12]. However, in the context of thermoelectric phenomena, the application of PINNs is still in its early stages, and existing studies only address selected aspects of the problem. For example, the work in [10] focuses on predicting temperature and voltage responses for different thermoelectric materials and identifying material properties as functions of temperature using neural-network-based inverse approaches. While valuable from a materials perspective, this study does not consider time-dependent operating conditions, such as oscillatory or dynamically varying currents, which are relevant for practical control and optimization scenarios.

The study in reference [11] investigates the transient behavior of a thermoelectric generator operating in energy-harvesting mode, i.e., converting a temperature difference into electrical power. However, the analysis is limited to the transient regime and does not address the quasi-steady operating phase. Additionally, it does not consider thermoelectric cooling systems driven by electrical energy input.

These limitations suggest the need for a unified PINN-based framework that can capture both the transient and quasi-steady regimes in electrically driven thermoelectric cooling systems and include realistic current excitation profiles. This study aims to address this gap by extending the PINN approach to model thermoelectric modules operating under time-dependent electrical currents, with a focus on temperature evolution relevant to practical cooling applications.

This work aims to investigate the possibility of using PINNs to solve equations that describe time-varying, transient, and quasi-steady thermoelectric phenomena in multistage systems. These systems consist of many cooperating, heat-conducting elements, and the model includes their characteristics. This work also aims to evaluate the effectiveness of this approach compared to traditional numerical methods.

2. Physics-Informed Neural Networks (PINNs)

The classic approach to using neural networks for mathematical modeling involves collecting a large amount of measurement data and teaching the network to identify relationships between inputs and outputs. In practice, however, this approach is often difficult to implement because it requires large training data sets, which are costly or impossible to obtain in many engineering applications.

The approach used in the PINN method is fundamentally different. The loss function design takes into account the residuals of the individual equations describing the physical model, calculated at selected points in the domain. These equations can be differential or algebraic. There are also examples of solving integral equations or equations containing fractional derivatives. Additionally, measurement points can be included in the loss function if available, but this is not a prerequisite. PINN is treated as a mathematical function u_{Θ} dependent on parameters Θ (weights), which transforms input data into output values through successive layers. There are several important differences compared to the training of classical neural networks. First, the activation functions must be differentiable, allowing the calculation of the appropriate derivatives using automatic differentiation algorithms. The loss function itself also plays a key role, containing terms that penalize the

failure to satisfy differential equations and boundary or initial conditions. In special cases, it is possible to determine the solution solely on the basis of differential equations and imposed conditions, without using any measurement data. Importantly, this approach does not require the construction of a computational grid, so PINN can be treated as a grid-free method to solve differential equations. As a result, the network is not only a universal approximator of input–output relationships, but also a tool for determining solutions that comply with the laws of physics, even with limited data availability.

To illustrate the idea, let us consider a simple network with one input, two hidden layers, and two outputs as shown in Figure 1.

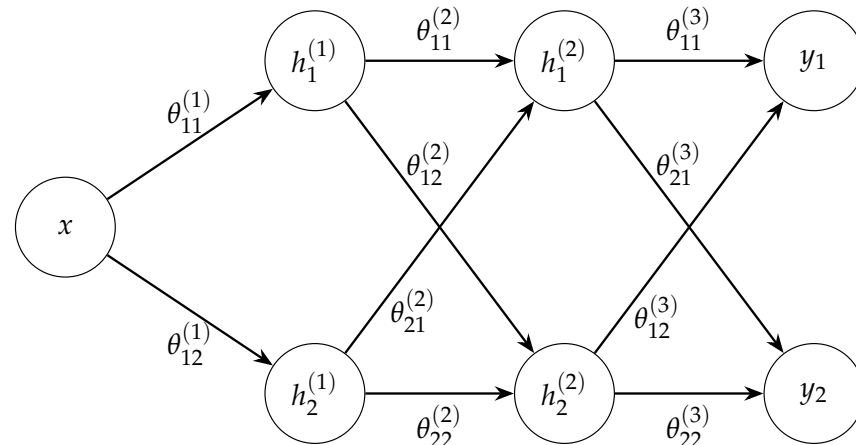


Figure 1. Diagram of a simple PINN network.

The physics-informed neural network is trained to approximate a vector-valued function $u(x) = (y_1(x), y_2(x))$, where each component is represented by a standard fully connected feedforward neural network composed of affine transformations and nonlinear activation functions.

The activation functions are selected in accordance with the physical characteristics of the modeled thermoelectric system, such as smoothness and stability of temperature evolution. This choice enables the network to represent time-dependent thermal responses while maintaining numerical robustness.

Since the network outputs are compositions of differentiable functions, they are differentiable with respect to both the input variable x and the network parameters Θ . The required derivatives are computed using automatic differentiation based on the chain rule, without resorting to numerical finite-difference approximations or explicit symbolic derivations.

This allows derivatives of arbitrary order to be incorporated directly into the loss function with machine precision, enabling a consistent enforcement of the governing differential equations and boundary conditions. As a result, the training process is explicitly guided by the physical laws describing the thermoelectric system, which constitutes a key feature of the PINN methodology.

Let us consider a system of differential equations describing the analyzed physical phenomenon, written in the form

$$\mathcal{R}(x, u(x), u'(x)) = 0,$$

where u is a vector of modeled quantities. In the PINN method, the solution $u(x)$ is approximated by a neural network $u_{\Theta}(x)$, and the degree to which the model equations are satisfied is evaluated by the residuals calculated at selected points in the domain.

In general, each residual has the form

$$\text{Res}_i(x, \Theta) = \mathcal{R}_i(x, u_\Theta(x), \partial_x u_\Theta(x)), i = 1, \dots, N_R$$

and measures the local deviation of the network approximation from satisfying the corresponding equation. The number of residuals N_R denotes the total number of residuals corresponding to the differential equations, the boundary conditions, and the algebraic relations that occur in the model.

Residuals corresponding to initial or boundary conditions and residuals resulting from additional algebraic relationships between network outputs are defined similarly.

Based on the defined residuals, a learning functional is constructed

$$\Phi_{\text{phys}}(\Theta) = \sum_{i=1}^{N_R} w_i \sum_{x_j \in \Omega_{\text{coll}}} |\text{Res}_i(x_j; \Theta)|^2,$$

where $\text{Res}_i(x_j; \Theta)$ denote the residuals corresponding to the differential equations, boundary conditions, or algebraic relations occurring in the model, $w_i > 0$ are the weights of individual components, while Ω_{coll} is the set of collocation points in the domain of computation. Minimizing the functional Φ_{phys} with respect to the network parameters Θ leads to an approximate solution consistent with the model equations and the imposed physical constraints.

An important feature of the PINN approach is that the learning process does not require access to large sets of measurement data. The primary sources of information here are equations describing the modeled phenomenon, while experimental data, if available, can be included in the learning functional as additional supervisory components.

Since the output of a neural network is a differentiable function with respect to the input variables, the derivatives appearing in the residues of differential equations can be determined consistently and accurately during the learning process, without the need for differential approximations.

3. Mathematical Model of Thermoelectric Module

The operation of the thermoelectric module results from the coupled Seebeck, Peltier, and heat conduction phenomena. In the task used for testing, a one-dimensional model of a system containing thermoelectric elements is adopted, in which the temperature distribution and heat fluxes are described by coupled differential equations.

The analyzed system is shown in Figure 2. It consists of an upper internal and lower heat reservoir and thermoelectric modules placed between them. The contact thermal resistance is taken into account at each junction between adjacent elements.

Thermoelectric modules are subject to phenomena related to the Seebeck, Peltier, and Joule effects, as well as heat conduction. Therefore, the corresponding heat fluxes can be described using the following equations:

$$\dot{Q}_u = k_1 (T_u - T_{u_{ir}}) + T_u \alpha_1 I_1(t) - \frac{1}{2} R_1 I_1(t)^2, \quad (1)$$

$$\dot{Q}_{u_{ir}} = k_1 (T_u - T_{u_{ir}}) + T_{u_{ir}} \alpha_1 I_1(t) + \frac{1}{2} R_1 I_1(t)^2, \quad (2)$$

$$\dot{Q}_{l_{ir}} = k_2 (T_{l_{ir}} - T_l) + T_{l_{ir}} \alpha_2 I_2(t) - \frac{1}{2} R_2 I_2(t)^2, \quad (3)$$

$$\dot{Q}_l = k_2 (T_{l_{ir}} - T_l) + T_l \alpha_2 I_2(t) + \frac{1}{2} R_2 I_2(t)^2, \quad (4)$$

where $\dot{Q}_u$ denotes the heat flux absorbed (cooling) by the module operating at a lower temperature (upper heat reservoir), $\dot{Q}_{u_{ir}}$ denotes the heat transferred by this module to the

internal heat reservoir, $\dot{Q}_{l_{ir}}$ is the heat absorbed from the internal reservoir by the module operating at a higher temperature, while $\dot{Q}_l$ describes the heat flux emitted by this module. The functions I_1 and I_2 denote the time-dependent waveforms of the supply currents to the first and second thermoelectric modules, respectively. Their form was adopted on the basis of earlier works [13,14], in which they were determined as a result of optimization procedures, and in this analysis they are treated as model input data.

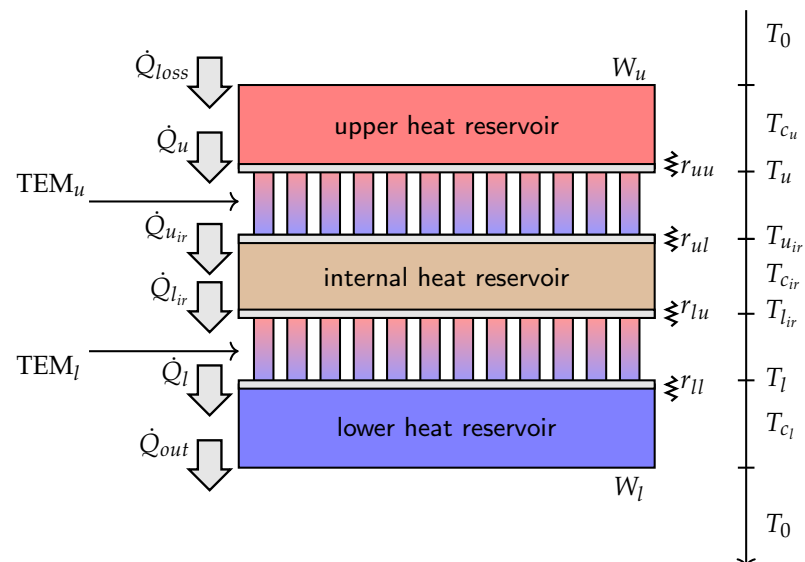


Figure 2. Diagram of a thermoelectric system. Red and blue colours illustrate the temperature gradient pictorially.

Furthermore, it was assumed that in the system under consideration, both thermoelectric modules are identical. It was assumed that the coefficients appearing in the above equations (in particular, the Seebeck coefficients α_1 and α_2 , the thermal conductivities k_1 , k_2 , and the electrical resistances R_1 , R_2) are independent of temperature. In all simulations presented in this work, the two thermoelectric modules are assumed to be identical, i.e.,

$$\alpha_1 = \alpha_2, \quad k_1 = k_2, \quad R_1 = R_2.$$

Separate parameter notation is retained for clarity and consistency with the general model formulation. Although identical parameter values are assumed, this notation facilitates interpretation of the model structure and its relation to the schematic shown in Figure 2.

Heat exchange with the environment is described by the following relationships:

$$\dot{Q}_{\text{loss}} = W_u \cdot (T_0 - T_{cu}), \quad (5)$$

$$\dot{Q}_{\text{out}} = W_l \cdot (T_{cl} - T_0), \quad (6)$$

where the first of the above equations describes the heat loss between the cooled tank and the environment, and it may also represent a form of process heat supplied to the cooling tank. The second equation similarly describes heat transfer to the environment on the hot side of the lower heat reservoir. It is assumed that the heat flux is directly proportional to the temperature difference in both equations.

In practice, the heat transfer coefficient W_l in the second equation is significantly greater than W_u , usually by one or several orders of magnitude. This is due to both the limited efficiency of thermoelectric technologies and the fact that the primary task of the system under consideration are cooling and effective heat dissipation.

The energy balance equations in each of the capacitive elements (upper, internal, and lower heat reservoirs) take the form

$$\dot{Q}_{\text{loss}} = \dot{Q}_u + c_u \frac{dT_{c_u}}{dt}, \quad (7)$$

$$\dot{Q}_{u_{ir}} = \dot{Q}_{l_{ir}} + c_{ir} \frac{dT_{c_{ir}}}{dt}, \quad (8)$$

$$\dot{Q}_{\text{out}} = \dot{Q}_l - c_l \frac{dT_{c_l}}{dt}, \quad (9)$$

where the above equations describe changes in the temperature of individual heat reservoirs over time. It is assumed that their heat capacities are constant and independent of temperature.

The relationships between heat flows and temperature differences at the junctions of thermoelectric modules with heat reservoirs were described using concentrated thermal resistances. It was assumed that the heat capacities of components other than reservoirs were negligible.

$$T_{c_u} - T_u = r_{uu} \dot{Q}_u, \quad (10)$$

$$T_{u_{ir}} - T_{c_{ir}} = r_{ul} \dot{Q}_{u_{ir}}, \quad (11)$$

$$T_{c_{ir}} - T_{l_{ir}} = r_{lu} \dot{Q}_{l_{ir}}, \quad (12)$$

$$T_l - T_{c_l} = r_{ll} \dot{Q}_l. \quad (13)$$

Equations (10)–(13) describe heat conduction through the thermal resistances of individual contacts, where the temperature difference between the elements is proportional to the heat flux flowing through a given contact.

In [13], it was noted that applying a suitably selected, optimized variable (periodic) current to thermoelectric modules allows one to achieve temperatures significantly lower than the minimum temperature possible in steady state. This effect results from the fact that a sudden increase in current immediately amplifies the Peltier effect, whereas the unfavorable heating of the whole system, especially heat sink, associated with the Joule effect increases with a certain delay resulting from the presence of heat capacities in the system. In two-stage configurations, the thermal inertia of the internal storage tank plays an additional role, maintaining a reduced temperature for a relatively long time and thus temporarily providing favorable operating conditions—a smaller temperature difference for the pumped heat—for both modules. A properly selected current waveform, in particular a periodic waveform, most often of pulsed or linearly increasing nature, allows the system to “overtake” the heating and achieve short-term cooling (supercooling), even by several Kelvins stronger than is possible under steady-state conditions.

The periodicity of the supplied current is essential here. It allows a very low temperature to be achieved at a selected point in the system by using its thermal inertia. This is made possible by using heat capacity with a relatively slow temperature change. This allows for an increase in Joule heat generated by the flow of current, while simultaneously increasing the Peltier effect at a lower operating temperature on both sides of the module. After reaching the minimum temperature, a certain amount of time must pass for the system to regain its cooling capacity. Therefore, once the minimum temperature has been reached, the current should be reduced or the pulse sequence repeated. This type of control allows for very effective use of the dynamic properties of Peltier modules, contrary to steady-state, where there is one single value of current for each module to reach maximum steady-state cooling performance (lowest temperature). However, this approach has some drawbacks. First, it can only be used in situations where periodic cooling is acceptable. It

also reduces the overall efficiency of the system. Furthermore, this phenomenon cannot be exploited at will—up to certain values, cooling improvement is achieved by extending the period, but at a certain point, no further improvement occurs [14].

As a result, the variable current acts as a transient control, allowing significantly lower temperatures to be achieved momentarily without the need to increase the size of the heat exchangers or modify the geometry of the system. It is therefore not only a practical tool but also an indispensable one in applications where fast, intense, and short-term cooling is important. The optimal current waveform is not intuitive and depends on complex, nonlinear interactions between the Peltier effect, Joule heating, thermal resistances, and heat capacities of individual layers. Therefore, it is necessary to precisely model the system dynamically to computationally find a current pulse shape that, depending on conditions such as the specified pulse period, target temperature, or geometry variability, maximizes supercooling while avoiding excessive heating of the system in the later phase.

The search for new tools and techniques to model thermoelectric phenomena remains an important research challenge. On the one hand, high model quality is crucial for the correct selection of the optimal current waveform and reliable prediction of the dynamics of phenomena. However, in practical applications, it is necessary to have a computational tool that combines high accuracy with computational efficiency and enables rapid determination of optimal controls, even for complex configurations of thermoelectric modules. Effective solvers can also be used for real-time control based on sensor data.

For these reasons, an attempt was made to evaluate the applicability of methods based on Physics-Informed Neural Networks, which enable the modeling of coupled physical phenomena without the need to simplify the equations describing the dynamics of the system. This approach allows for the preservation of a continuous time description and consistency with the laws of physics, which is particularly important in the analysis of transient phenomena.

Classic differential models, despite their physical correctness, often involve high computational costs and limited scalability, especially in the case of compact and integrated cooling systems, where precise mapping of geometry and boundary conditions is necessary. In this context, PINN-based methods can be an alternative computational tool, combining explicit enforcement of compliance with physical equations with the flexibility characteristic of machine learning methods. After the training stage, they enable rapid prediction of system behavior, which can significantly reduce computational cost in parametric analyses, multiple simulations, and real-time control. This creates potential opportunities for using PINN models to quickly optimize the operation of circuits in embedded systems that respond to current measurement signals.

Achieving comparable accuracy with less computational overhead makes it possible to use such models to analyze the impact of current waveforms and control strategies in thermoelectric modules, including the study of transient phenomena associated with periodic current control, such as the supercooling effect.

Modeling Using PINN

In order to assess the applicability of the PINN method for describing temperature dynamics in a thermoelectric module, an approach based on minimizing a physical functional constructed from the model equations is considered. This functional takes into account both the energy balance equations and the relationships that describe the heat flow in individual parts of the thermoelectric module.

In particular, the physical functional takes the form

$$\Phi_{\text{phys}} = \sum_{i=1}^7 w_i \sum_{t_j \in \Omega_{\text{coll}}} |\text{Res} R_i(T(t_j), \dot{T}(t_j))|^2,$$

where R_i , $i = 1, \dots, 7$, denote the residuals corresponding to the physical relations included in the model (Equations (7)–(13)), including energy balances and relationships describing the heat flow in successive parts of the thermoelectric module, as well as balance equations supplementing the model. Each residual measures the local degree of non-compliance with the corresponding physical relation by approximating the temperature $T(t)$ generated by the neural network. The coefficients $w_i > 0$ allow adjustment of their relative influence in the learning process. The set Ω_{coll} denotes collocation points in the time domain. Minimizing the functional Φ_{phys} with respect to the network parameters is the basis for training PINNs. The disappearance of all residuals is equivalent to satisfying the system of equations under consideration, while in the process of training a PINN, the aim is to minimize them in the mean square sense, according to the definition of the functional Φ_{phys} .

In addition, the model takes into account the initial conditions

$$T_k(0) = T_k^{(0)}, \quad k = 1, \dots, 7, \quad (14)$$

The corresponding temperatures in Figure 2 are presented in Table 1. They are also treated as constraints in the learning process. For this purpose, they are introduced into the cost function in the form of additional penalty terms

$$(T_k(0) - T_k^{(0)})^2.$$

Taking into account both the model equation residuals and the initial conditions leads to a complete system of constraints used in training the PINN. The learning process consists of determining the network parameters Θ for which the value of the functional Φ_{phys} is as close to zero as possible.

Table 1. State variables (temperatures) predicted by the PINN and their correspondence to physical nodes shown in Figure 2.

PINN Output	Temperature Symbol	Physical Node (Figure 2)
1	T_u	upper side upper module
2	$T_{u_{ir}}$	lower side upper module
3	$T_{c_{ir}}$	internal heat reservoir
4	$T_{l_{ir}}$	upper side lower module
5	T_l	lower side lower module
6	T_{c_u}	upper heat reservoir
7	T_{c_l}	lower heat reservoir

4. Parameter Selection, Sensitivity Analysis, and Learning Methodology

In the implementation of a PINN, the key elements are the selection of weights for individual components of the loss function Φ_{phys} , the strategy for selecting collocation points, and the organization of the learning process. These elements directly affect the stability of optimization, the network's ability to map the temporal dynamics of the system, and the correctness of the physical relations imposed by the model. All model equations, both differential and algebraic, are used in the learning process in the form of appropriate residuals in the loss function, with weights assigned to them reflecting their relative importance in the optimization process.

This chapter discusses the selection of loss function weights, the sensitivity analysis of the PINN solution to their change, and the adopted learning methodology, with particular emphasis on the strategy of distributing collocation points over time and the course of the learning curve.

4.1. Selection of Loss Function Weights and Sensitivity Analysis

The energy balance equations described by relations (7)–(9) are ordinary differential equations and describe the evolution of temperatures over time in the thermal capacities of the system. They are responsible for the correct mapping of heat flow dynamics, solution stability, and system behavior over a longer time horizon. For this reason, the compatibility of the PINN solution with these equations is crucial from the point of view of the physics of the modeled process.

Algebraic Equations (10)–(13) describe the local relationships between heat fluxes and temperature differences resulting from the adopted thermal resistance model. They ensure the local physical consistency of the solution, but do not directly determine its evolution over time.

At the initial stage of the research, it was noticed that equal weights of individual residuals in the loss function cause the network to underlearn. It was decided to vary the weights according to which equations proved easier for the network. It turned out that weights of 0.2 for algebraic equations and 1.0 for differential equations work well. The sensitivity analysis discussed below also supports this approach.

In order to assess the impact of the choice of loss function weights, a sensitivity analysis of the PINN solution to changes in the relative weights of differential and algebraic equations was performed. Several configurations were considered, in which the weights of algebraic equations were changed in the range 0.1–0.3, while the weights of differential equations were changed in the range 0.9–1.1. For each configuration, five independent learning processes were performed, and the average value of the loss function on the training set was taken as a measure of quality.

The lowest average value of the loss function was obtained for weights equal to 0.2 for algebraic equations and 1.0 for differential equations. Both decreasing and increasing these weights led to an increase in the loss function value, indicating the existence of an optimal ratio between the components of the loss function. The differences between the analyzed configurations were quantitative in nature, while maintaining a similar quality of temperature curve mapping.

The adopted weight ratio, in which the weights of algebraic equations constitute approximately 20% of the weights of differential equations, provides a compromise between the correct mapping of the time dynamics of the system and the preservation of physical consistency resulting from algebraic relations.

4.2. Training Strategy and Selection of Collocation Points

The convergence analysis of the learning process was conducted in the time interval $[0, b]$ s, where

$$b = \min(2500, 5\tau), \quad (15)$$

which allows for the evaluation of both the transient phase and the quasi-steady state, as well as comparison with the reference solution.

Numerical studies have shown that including too long a segment corresponding to the quasi-steady state in the learning process leads to a deterioration in the quality of the initial phase mapping. In such a situation, the number of collocation points enforcing the model equations in the steady state significantly exceeds the number of points representing the start-up phase, as a result of which the loss function minimization process favors the

matching of the system behavior after stabilization at the expense of the correct mapping of the initial dynamics.

Therefore, from the beginning of the learning process, an uneven distribution of collocation points in the time domain was used. In particular, 2000 collocation points were distributed in the interval $[0, 2\tau]$, corresponding to the transition phase characterized by the fastest temperature changes, while the remaining 1000 points were distributed in the interval $[2\tau, b]$.

The strategy used to select collocation points counteracts the above-mentioned effect by increasing the relative influence of the transient phase on the value of the loss function and preventing the premature dominance of the quasi-steady-state solution. As a result, the PINN is forced to simultaneously map the initial dynamics and the behavior of the system over a long time horizon.

4.3. Learning Curve and Stability of the Optimization Process

An example learning curve is shown in Figure 3 in the form of the logarithm of the loss function value in successive iterations of the learning process. A general downward trend is visible, indicating the convergence of the optimization process, but the decrease in the loss function is not monotonic.

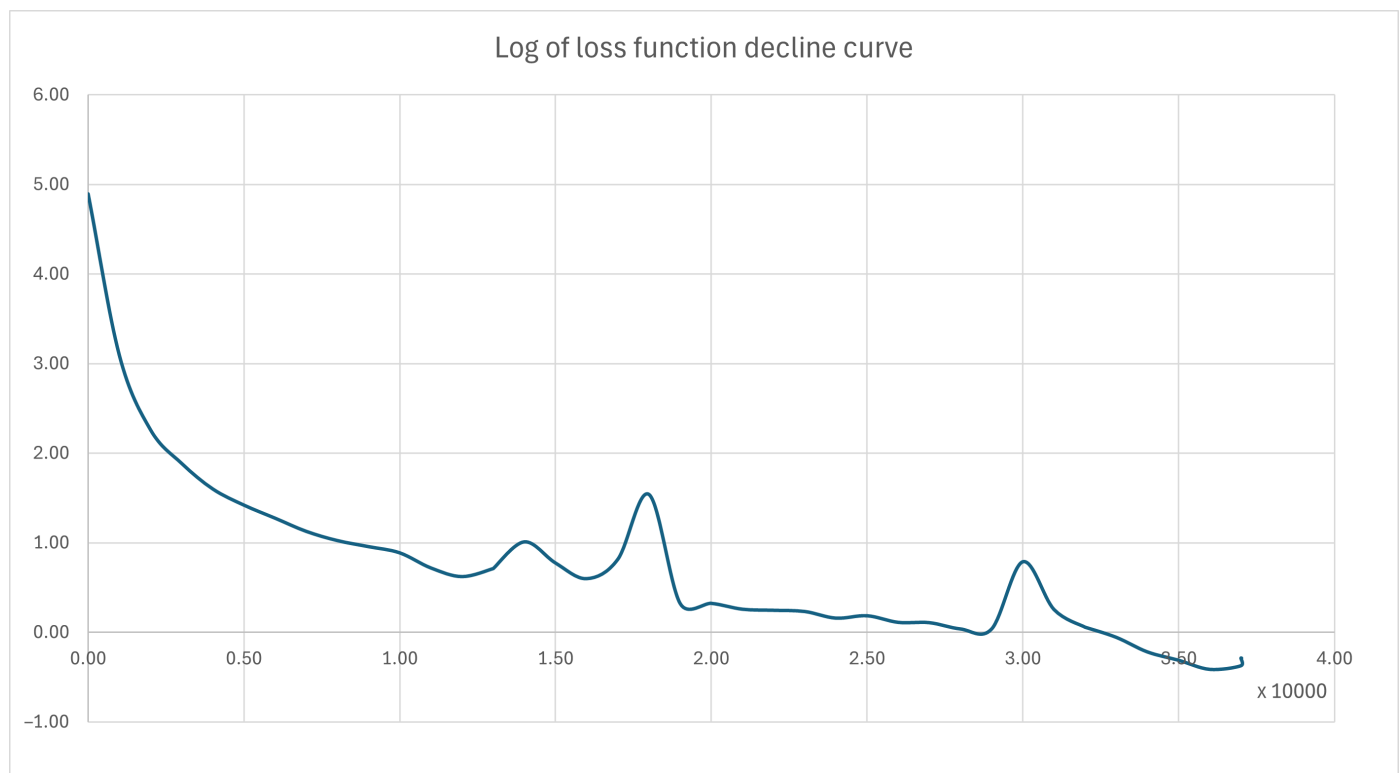


Figure 3. Log of the loss function in a typical example.

The non-uniform course of the learning curve results from the resampling technique used for collocation points. The PINN is not trained on a fixed set of points, but every 1000 iterations a new subset of collocation points is randomly selected from a fixed time distribution. This significantly reduces the computational cost of the learning process, but causes temporary increases in the value of the loss function, visible in the graph as local maxima.

This behavior of the learning curve is characteristic of methods based on random sampling of the domain and does not indicate instability of the optimization process. On the contrary, the observed trend indicates a gradual increase in the network's ability to satisfy

the model equations across the entire time domain, despite periodic changes in the set of collocation points.

The resampling strategy used, combined with the uneven distribution of collocation points over time, promotes the correct mapping of both the transient phase and the quasi-steady state and prevents excessive fitting of the network to a specific set of training points.

5. Results

The PINN used in the calculations had a multi-layer architecture which, after appropriate selection of hyperparameters, proved sufficient to reproduce both the transient state (the initial phase of the process) and the quasi-steady state achieved usually after several periods of excitation. The network structure, shown in Figure 4, includes an input layer, a Fourier layer, two hidden layers, and an output layer generating seven temperature components.

The weights of the network were initialized using the Glorot uniform method [15], while the hyperbolic function $\tanh$ was used as the activation function. The learning process was carried out using the Adam optimizer [16], with an initial learning rate of 5×10^{-4} , which was reduced to 10^{-4} in the later stages of training. In the final stage of optimization, the network parameters were further adjusted using the *L-BFGS* [17].

During training, different weights were applied to the physical functional components corresponding to the individual equations of the model. The algebraic Equations (10)–(13), considered less numerically demanding, were assigned a weight of $w_i = 0.2$. The differential Equations (7)–(9) were given higher weights $w_i = 1$, to strengthen their impact on the learning process.

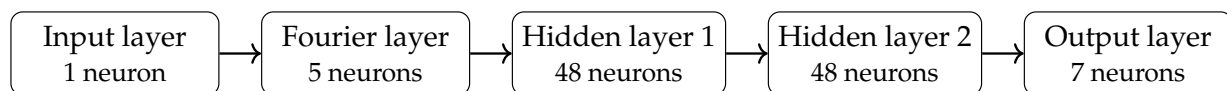


Figure 4. Diagram of the neural network.

Numerical tests have shown that the system under consideration is characterized by a high degree of computational difficulty and, in this form, poses a significant challenge to the standard PINN architecture described above. In particular, these difficulties are related to the simultaneous mapping of the initial phase (transient state) and quasi-steady-state behavior while maintaining the stability of the learning process. For this reason, it was necessary to introduce modifications that involve explicitly imposing some of the model constraints and enriching the representation of the network input. The techniques used are listed below.

1. Hard boundary conditions [18]. In order to improve the convergence of the learning process with the loss function, the components corresponding to the initial conditions (14) were removed. Instead, their fulfillment was explicitly enforced by an appropriate construction of the approximation function. The desired temperature approximation was defined as

$$\bar{T}(t) = \frac{t}{t+1} NN(t) + \bar{T}_0, \quad (16)$$

where $NN(t)$ denotes the output of the neural network, while $\bar{T}_0$ is a vector of initial temperature values. This form ensures that the initial conditions are automatically satisfied because for $t = 0$, $\bar{T}(0) = \bar{T}_0$, regardless of the form of the function $NN(t)$. In the case under consideration, the time t is expressed in milliseconds; therefore, the normalization factor

$$\frac{t}{t+1} \quad (17)$$

takes the value zero at the initial moment and very quickly approaches unity for larger values of t , without significantly affecting the asymptotic behavior of the solution. The use of hard boundary conditions eliminates the need for the neural network to learn the initial conditions, which in practice leads to increased solution accuracy and accelerated training.

2. Fourier features [19]. The analyzed phenomenon is controlled by a periodically varying current, which leads to the occurrence of two characteristic phases: the initial transient regime and the quasi-steady state. In the latter case, periodic behavior can be expected, with a period equal to the excitation period.

In order to facilitate the representation of such relationships in the neural network, not only the time variable t was used as input, but also its periodic transformations. The input vector takes the form

$$x(t) = \begin{bmatrix} t \\ \sin(2\pi\omega t) \\ \cos(2\pi\omega t) \\ \sin(4\pi\omega t) \\ \cos(4\pi\omega t) \end{bmatrix}. \quad (18)$$

The use of Fourier features enables the network to capture the periodic components of the solution more effectively, particularly in the quasi-steady-state phase. Numerical tests have shown that omitting this modification leads to a correct reconstruction of the transient state preceding the appearance of repetitive pulsations characteristic of the quasi-steady phase, but results in a significant deterioration in the quality of the temperature reconstruction over a longer time horizon.

Taking into account the above modifications, the PINN training process required approximately 3×10^4 iterations.

The results of the proposed algorithm indicate its potential for application. The adopted test procedure consisted of comparing the solutions obtained with the numerical–analytical solution determined in the Wolfram Mathematica package.

Consequently, for the correct reproduction of the transient phase, it is crucial to conduct modeling within an appropriately selected, limited time interval.

For testing purposes, it was assumed that the current waveforms appearing in Equations (1)–(4) have the following form:

$$I_i(t) = I_i^{\min} + (I_i^{\max} - I_i^{\min}) \cdot \frac{(t \bmod \tau)}{\tau}$$

respectively:

- I_1 , a piecewise linear periodic (sawtooth) current waveform increasing from $I_1^{\min} = 3$ A to $I_1^{\max} = 4$ A over a segment of length τ ,
- I_2 , a piecewise linear periodic (sawtooth) current waveform increasing from $I_2^{\min} = 7$ A to $I_2^{\max} = 10$ A over a segment of length τ .

The adopted piecewise linear periodic current profiles were selected to ensure sufficient excitation of the system and to allow the PINN model to capture various thermoelectric effects under time-varying operating conditions. The current sawtooth used in the tests were treated as input data in this analysis. The numerical values for the parameters are given in Table 2.

To test the method, various values of the forcing period $\tau \in \{300, 500, 700, 900\}$ ms and various values of the heat capacities c_u , c_{ir} and c_l were analyzed.

The built-in numerical solver available in the Wolfram Mathematica environment was used as the reference solution for the system of coupled ordinary differential equations under consideration. This solver uses adaptive time integration methods, including both Runge–Kutta schemes and backward difference methods (BDF), automatically selected depending on the numerical properties of the system, in particular its potential stiffness. The solver used is part of a professional computing platform, widely used for both analytical and numerical analyses, which further justifies its selection as a reference solution.

The time histories of state variables obtained in this way are characterized by high numerical accuracy and constitute a baseline solution for validating the results obtained using the PINN. In particular, the ODE solution is used to generate reference temperature profiles, on the basis of which error measures are calculated and visual comparisons of both approaches are made.

Figures 5–12 show example temperature curves obtained using the PINN method and the reference solution obtained using a classical numerical ODE solver implemented in Wolfram Mathematica for selected values of the forcing period τ , as well as for different values of parameters c_{ir} and c_l . A comparison of the curves allows for a qualitative assessment of the consistency of the solutions in the transient phase and in the quasi-steady state.

Table 2. Symbols in Equations (1)–(13).

Symbol	Value	Unit	Meaning
k_1 k_2	2.1	$\frac{W}{K}$	Module thermal conductivity
α_1 α_2	0.08	$\frac{V}{K}$	Seebeck coefficients
R_1 R_2	1.587	Ω	Electrical resistance
r_{uu} r_{ul} r_{lu} r_{ll}	0.05	$\frac{K}{W}$	Contact thermal resistance
T_0	293.0	K	Ambient temperature
W_u W_l	0.5 10	$\frac{W}{K}$	Heat transfer coefficients

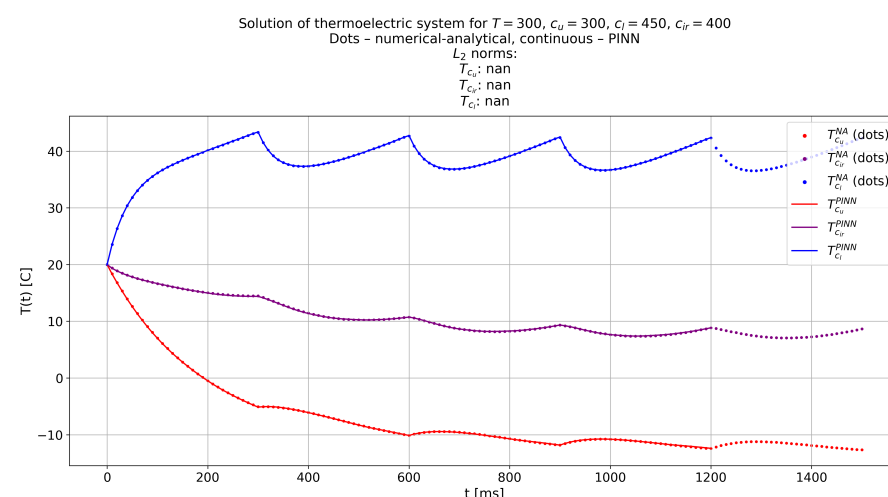


Figure 5. $\tau = 300$ ms, $c_u = 300$ JK^{−1}, $c_l = 450$ JK^{−1}, $c_{ir} = 400$ JK^{−1}.

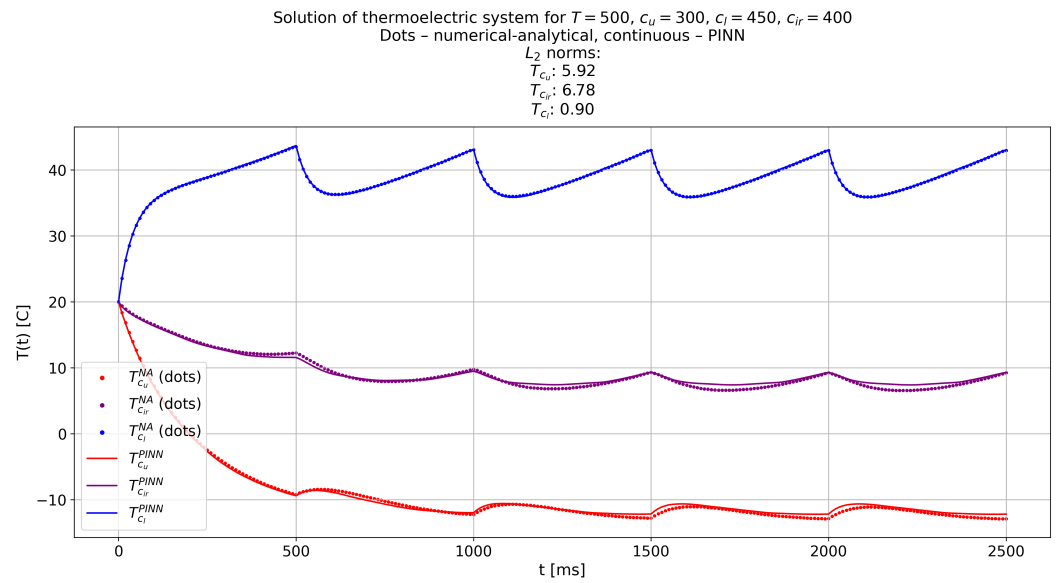


Figure 6. $\tau = 500$ ms, $c_u = 300 \text{ JK}^{-1}$, $c_l = 450 \text{ JK}^{-1}$, $c_{ir} = 400 \text{ JK}^{-1}$.

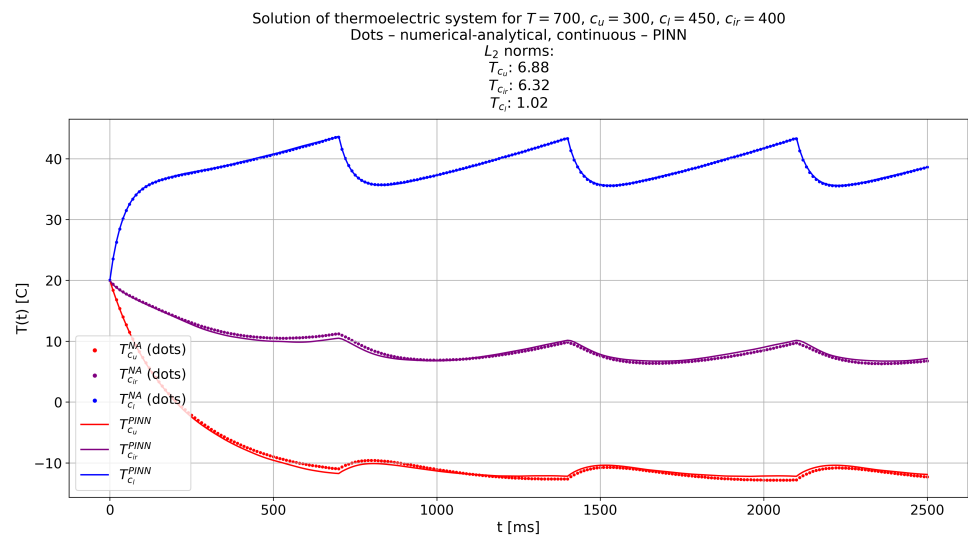


Figure 7. $\tau = 700$ ms, $c_u = 300 \text{ JK}^{-1}$, $c_l = 450 \text{ JK}^{-1}$, $c_{ir} = 400 \text{ JK}^{-1}$.

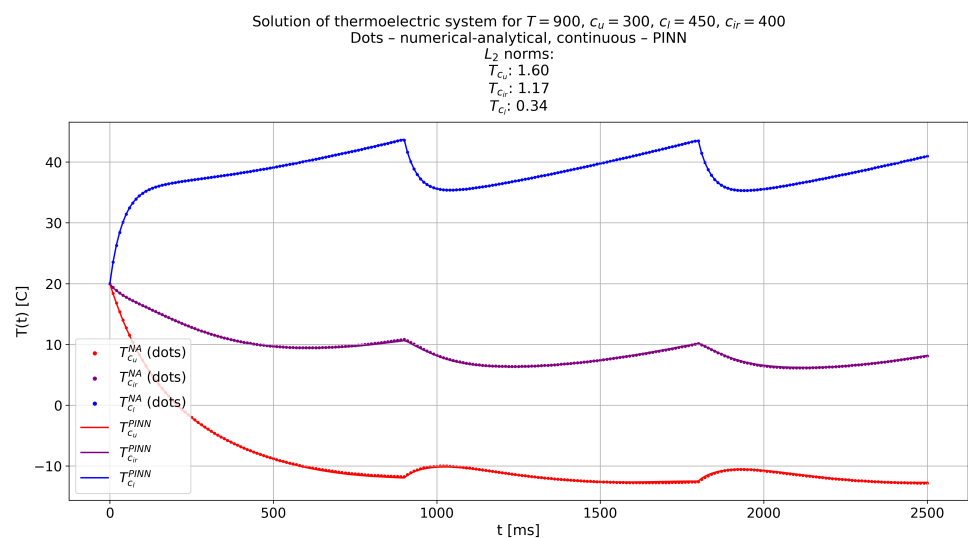


Figure 8. $\tau = 900$ ms, $c_u = 300 \text{ JK}^{-1}$, $c_l = 450 \text{ JK}^{-1}$, $c_{ir} = 400 \text{ JK}^{-1}$.

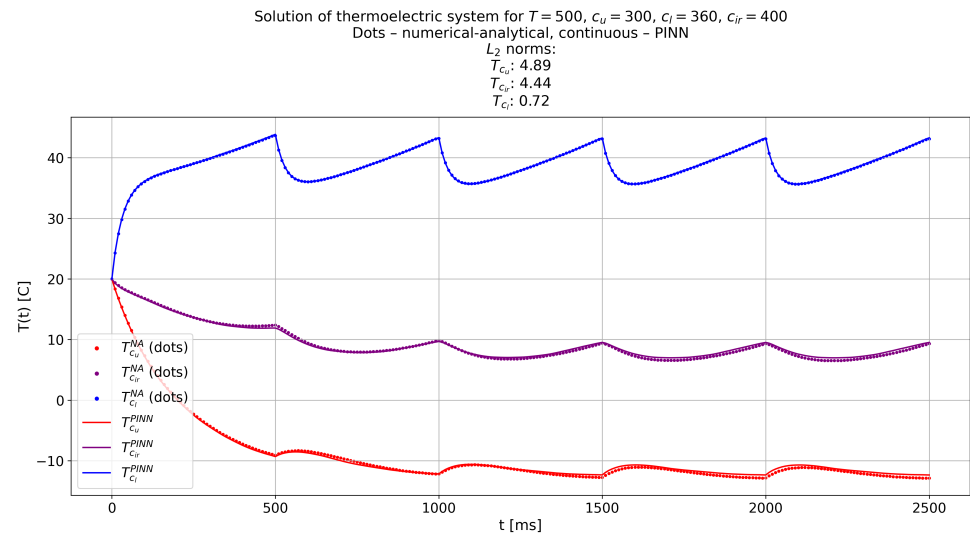


Figure 9. $\tau = 500$ ms, $c_u = 300 \text{ JK}^{-1}$, $c_l = 360 \text{ JK}^{-1}$, $c_{ir} = 400 \text{ JK}^{-1}$.

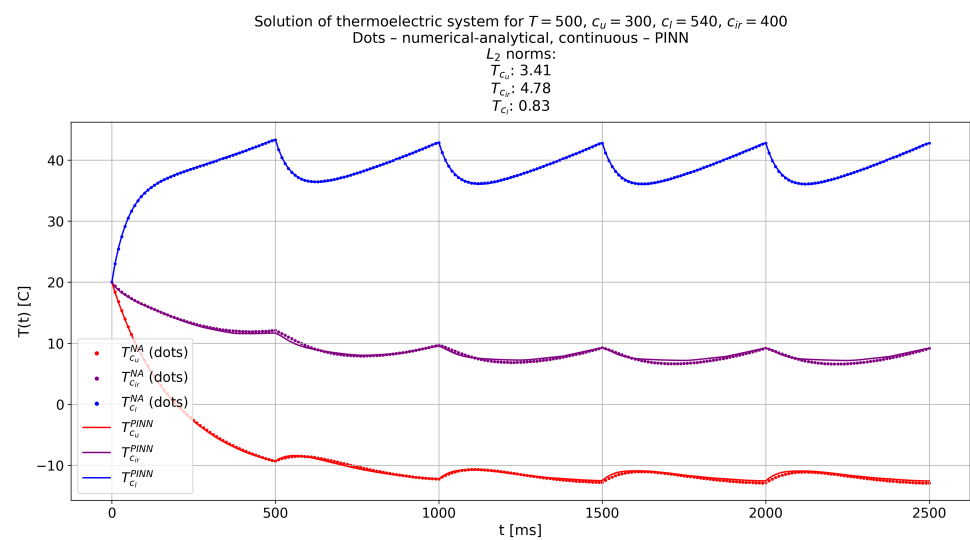


Figure 10. $\tau = 500$ ms, $c_u = 300 \text{ JK}^{-1}$, $c_l = 540 \text{ JK}^{-1}$, $c_{ir} = 400 \text{ JK}^{-1}$.

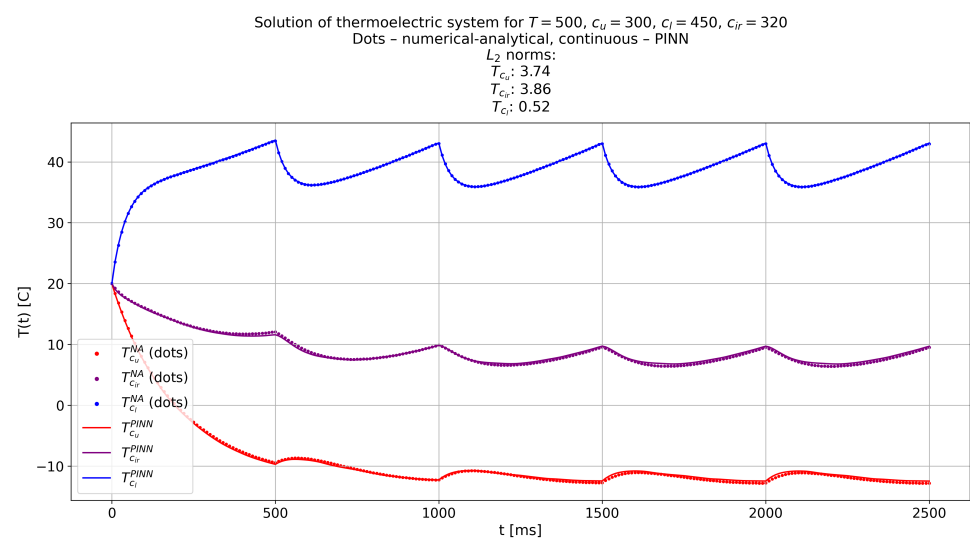


Figure 11. $\tau = 500$ ms, $c_u = 300 \text{ JK}^{-1}$, $c_l = 450 \text{ JK}^{-1}$, $c_{ir} = 320 \text{ JK}^{-1}$.

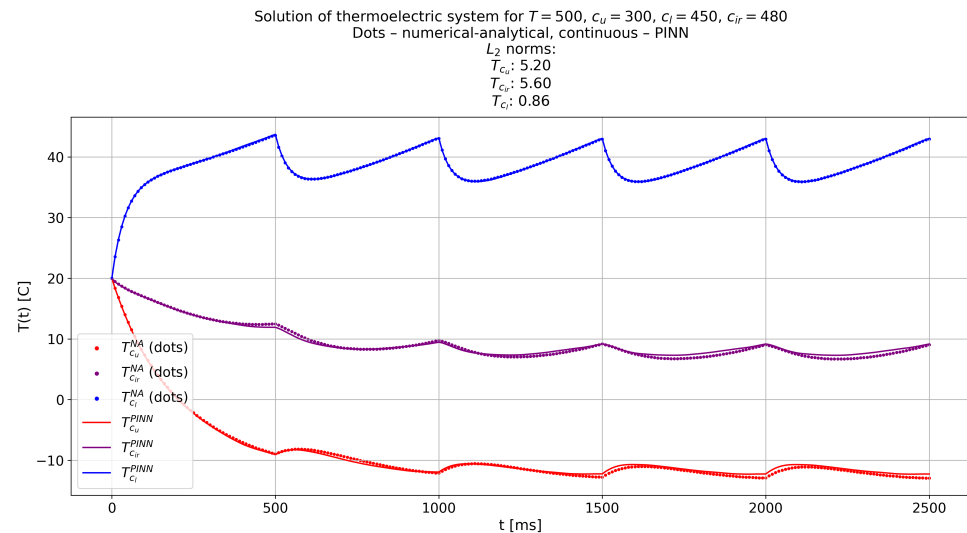


Figure 12. $\tau = 500$ ms, $c_u = 300$ JK^{−1}, $c_l = 450$ JK^{−1}, $c_{ir} = 480$ JK^{−1}.

The plots of observed temperatures show good agreement between the solutions obtained using the PINN method and the numerical-analytical method in a quasi-steady state. The greatest differences are evident in the initial start-up phase, particularly for shorter forcing periods. In this phase, discrepancies in temperature change dynamics are most pronounced, whereas after reaching a quasi-steady state, the obtained curves obtained remain consistent in terms of both quality (shape) and quantity (values). In order to quantitatively evaluate the results obtained for each of the physical temperatures analyzed, a reproduction error measure was introduced, defined using the L_2 norm of the deviation between the solution obtained by the PINN method and the numerical-analytical solution. This error takes the form

$$E_* = \|T_*^{\text{PINN}} - T_*^{\text{NA}}\|_2 = \sqrt{\sum_{i=1}^n (T_*^{\text{PINN}}(t_i) - T_*^{\text{NA}}(t_i))^2}, \quad (19)$$

where $*$ $\in \{c_u, c_{ir}, c_l\}$, while t_i , $i = 1, \dots, n$, denote a discrete set of time moments at which the solutions were compared. Table 3 reports error measures (RMSE and maximum error) only for the three temperatures T_{cu} , T_{cl} , and T_{ci} , which are the primary observable quantities directly related to the performance and efficiency of the thermoelectric system and therefore constitute the main targets of quantitative validation of the PINN model. The remaining temperatures act as internal state variables and are included to ensure physical consistency of the model, but are not evaluated separately in the error analysis.

An analysis of the error values summarized in Table 3 shows that the model is particularly sensitive to variations in the parameter c_{ir} , which lead to the largest deviations from the reference numerical solution.

Table 3. Numerical results.

τ	c_u	c_l	c_{ir}	E_{cu}	E_{cl}	E_{ci}
300	300	450	400	1.44	1.84	0.42
	300	450	480	4.83	4.05	0.81
	300	540	400	1.48	1.52	0.43
	360	450	400	2.55	2.09	0.50

Table 3. *Cont.*

τ	c_u	c_l	c_{ir}	E_{c_u}	E_{c_l}	$E_{c_{ir}}$
500	300	450	400	5.92	6.78	0.90
	300	450	320	3.74	3.86	0.52
	300	450	480	5.20	5.60	0.86
	300	360	400	4.90	4.44	0.72
	300	500	400	5.80	5.33	0.63
	300	540	400	3.41	4.78	0.83
	240	450	400	3.34	3.09	0.70
	360	450	400	3.84	3.27	0.54
700	300	450	400	6.88	6.32	1.02
	300	450	480	3.90	4.69	0.68
	300	540	400	2.63	2.27	0.47
	360	450	400	6.98	6.68	0.97
900	300	450	400	1.60	1.17	0.34
	300	450	480	11.16	10.67	1.57
	300	540	400	1.66	1.70	0.37
	360	450	400	8.87	7.45	1.04

6. Conclusions

The paper analyzes the possibility of using Physics-Informed Neural Networks to model temperature dynamics in a system with thermoelectric modules described by a system of coupled differential equations. Numerical tests confirmed that the PINN approach allows for obtaining approximate solutions that are in good agreement with the numerical–analytical solution, both in the quasi-steady state and, with an appropriate choice of time horizon, in the transition phase preceding the achievement of repeatable quasi-steady-state pulsations.

Significant advantages of the method include the ability to determine the solution in continuous form, without the need to introduce a computational grid, as well as the natural integration of differential equations, initial conditions, and algebraic relationships within a single learning functional. This method also allows the inclusion of measurement data as additional supervisory constraints, making it a flexible tool for engineering applications. However, the experiments revealed some limitations of the PINN approach, in particular the relatively long training time of the network and the sensitivity of the learning process to the choice of hyperparameters, functional weights, and the length of the analyzed time interval. It has also been shown that excessive dominance of the quasi-steady state in the set of collocation points can lead to a deterioration in the mapping of the start-up phase, which requires conscious design of the learning procedure.

Regardless of the pulsation period adopted for a given case, the simulated time should correspond to the length of several reference time periods for which significant supercooling gain relative to the steady state occurs. This range is closely related to the time constant of the system, resulting from the thermal efficiency of the thermoelectric modules and the heat capacities of the system components. Periods significantly shorter than the characteristic response time of the system do not bring significant benefits compared to constant current, while further increasing the pulse period above the values resulting from the physical parameters of the system does not lead to substantial changes in the achievable temperature in the quasi-steady state. In the analyzed case, the reasonable range of excitation periods is between approximately 50 and 2000 s. This observation has a direct impact on the selection of the simulation time horizon in the PINN approach, indicating that its length should be adjusted to the characteristic dynamic times of the

system. This paper analyzes cases with different forcing periods while maintaining the same simulation time horizon, which enabled a direct comparison of the quality of the reproduced waveforms within a single numerical experiment.

Further research plans include analyzing the possibility of reducing computational costs, e.g., by optimizing the network architecture, adaptively sampling collocation points, and modifying the weighting strategy of the functional components. Another interesting direction of development is the assessment of the usefulness of trained PINN models for rapid prediction of the behavior of thermoelectric modules in real-time applications, including embedded systems and control systems.

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